Data Mining-- Classification

Business Analytics,1e

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Chapter 8 Learning Objectives (LOs)

LO 8.1 Describe the data mining process.

LO 8.2 Implement kNN as an example of Classification.

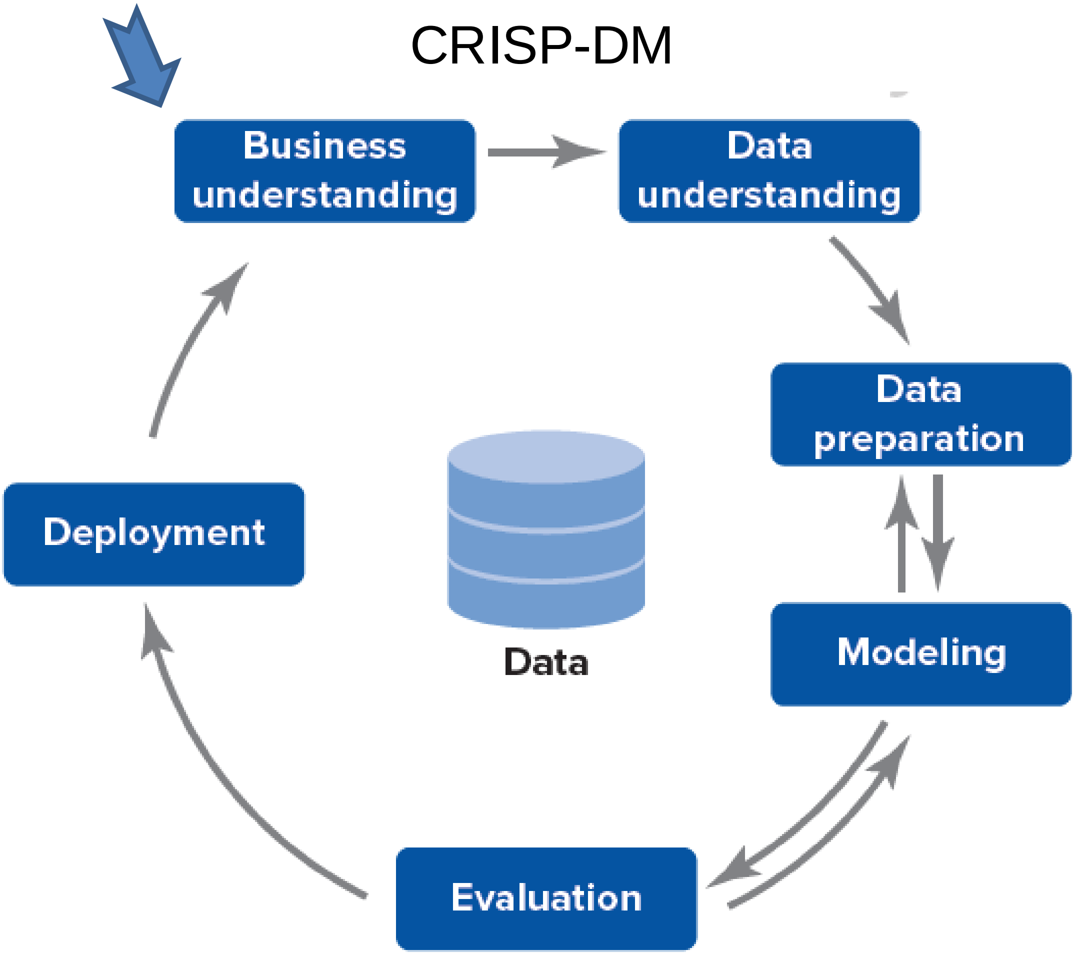
LO 8.3 Assess the predictive performance of data mining models.

Data Mining Overview

- Data mining is a complex process of examining large sets of data for identifying patterns and then using them for gaining business insights.
- Established standards with defined steps
  - CRISP-DM: Cross-Industry Standard Process for Data Mining
  - SEMMA: Sample, Explore, Modify, Model, and Assess
- CRISP-DM found to be more popular
  - Usually, 80% of time spent on data mining process is devoted to data understanding + preparation

Data Mining Overview

CRISP-DM



Data Mining Overview

SEMMA

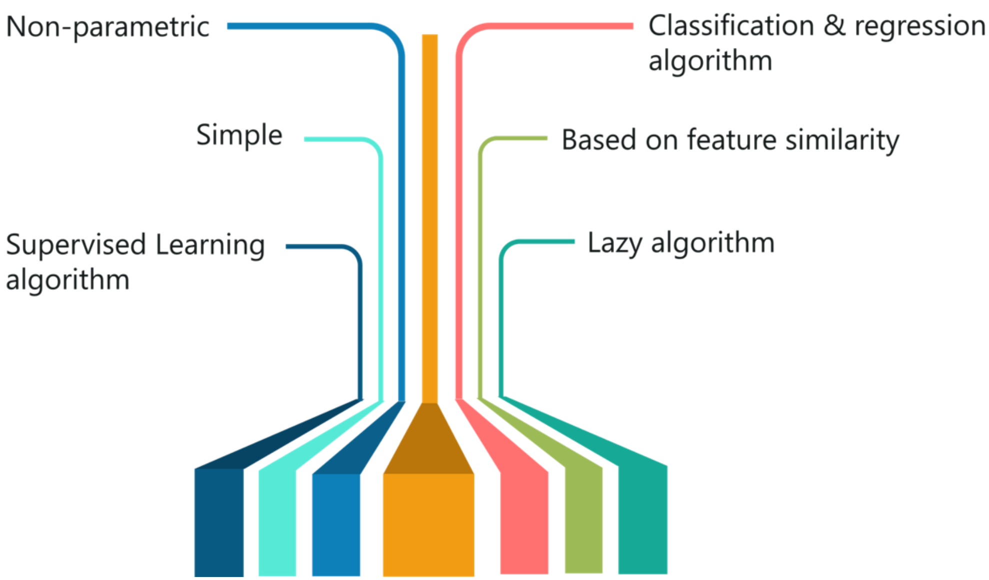
- Sample: identify appropriate variables, merging and/or dividing, sample
- Explore: exploratory data analysis
- ***Modify***: variables are selected, created, and/or transformed
- Model: analysis techniques and models are chosen and applied
- Assess: results from different models are presented to end users, compare outcomes and performance, new observations are scored

Supervised data mining,

- Classification model: target is categorical, objective is to predict class
  - example:
    - For an incoming email, classify it as business, personal, junk, phishing etc.
    - For stock trading: identify (classify) buy, hold, or sale action based on the given state of conditions
- Regression model: target is numeric, objective is to predict the target for a new case
  - example:
    - spending of a customer in each month
    - For stocking trading, forecast the index/price at the year-end.

The *k*-Nearest Neighbors Method

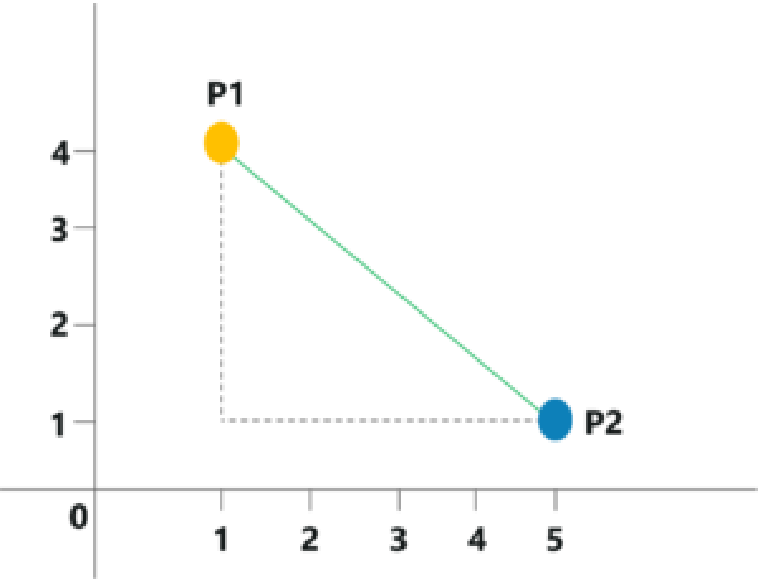
- The *k*-nearest neighbors (KNN) method is a memory-based reasoning technique (lazy algorithm).
  - Simple yet effective
  - Frequently used
  - Produces acceptable results
  - Time consuming and computationally expensive
- It uses observations from the past that are the most similar to new observations to classify or predict the target variable.
- It does not assume an underlying distribution (nonparametric).



The *k*-Nearest Neighbors Method

- Traditional methods require the predictor variables to be numerical:
  - transform categorical variables to numeric.
- KNN identifies a number of existing observations that are most similar to the new observation for classification.
- There are two components.
  1. Similarity measure (typically Euclidean distance)
  2. Number of nearest neighbors to consider, *k*
- Use data partitioning for Training and Validation sets.
  - Search the entire training data set for the most similar observation, provide a score for the new observation.
  - Experiment with a range of values for *k*

Algorithm of kNN



Point P1 = (1,4)

Point P2 = (5,1)

Euclidian distance = $\sqrt{(5-1)^2+(4-1)^2} = 5$

The simple algorithm of k-NN for classifying items:

1. Specify *k* value (number of nearest neighbors).

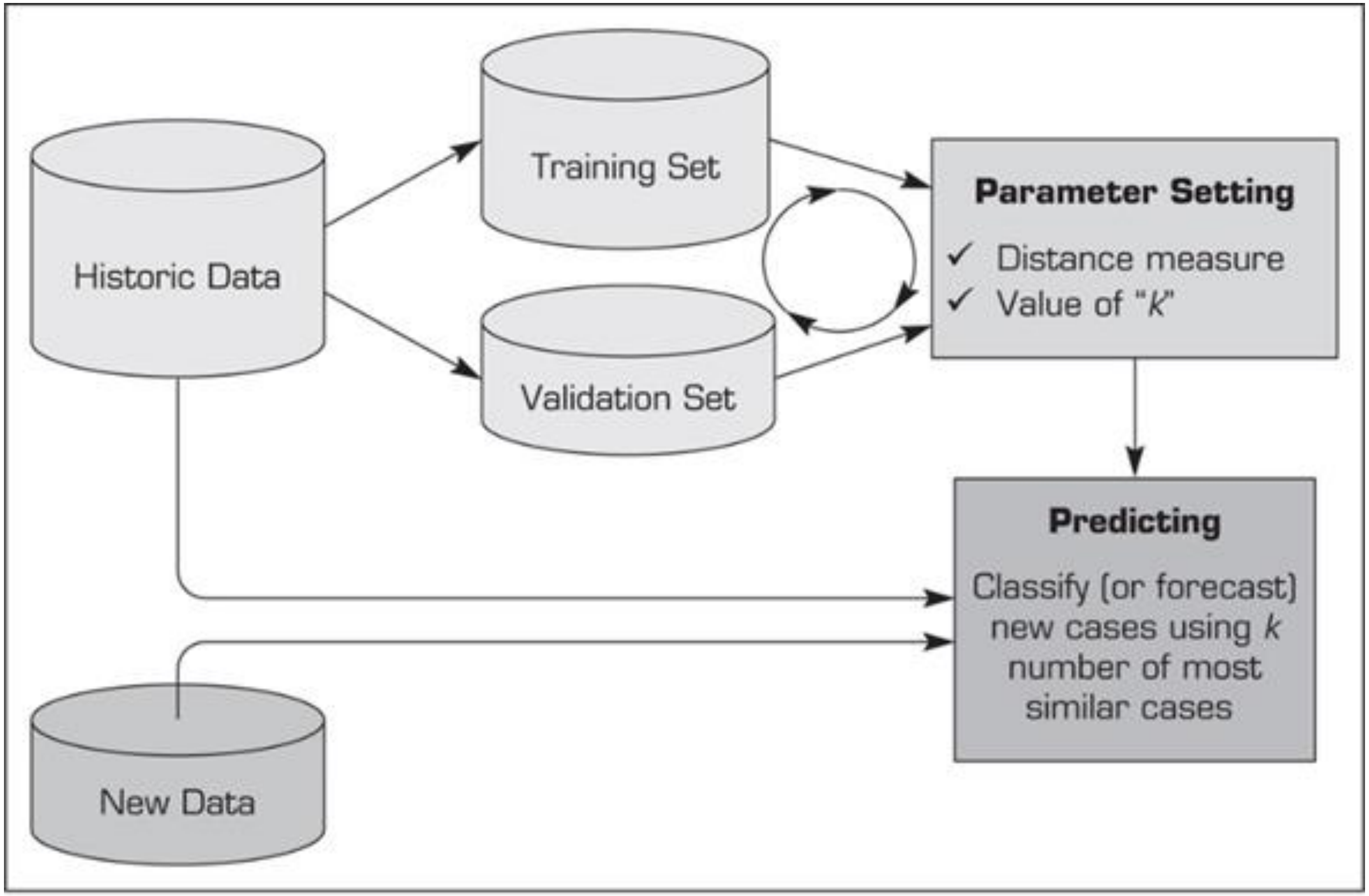
2. Compute distances between the query-instance and all the training instances, e.g. Euclidean distance.

$$d(i,j) = \sqrt[q]{\left(\left|x_{i1}-x_{j1}\right|^q + \left|x_{i2}-x_{j2}\right|^q + \cdots + \left|x_{ip}-x_{jp}\right|^q\right)}$$

3. Determine k nearest neighbors (the comparative distances) using the sorted distance obtained in (2)

4. Determine the class of the query-instance using the majority vote technique (outnumber of the classes of the nearest neighbors)

kNN process



Training Set
Historic Data
Validation Set
Parameter Setting
✓ Distance measure
✓ Value of "k"
Predicting
Classify (or forecast) new cases using k number of most similar cases
New Data

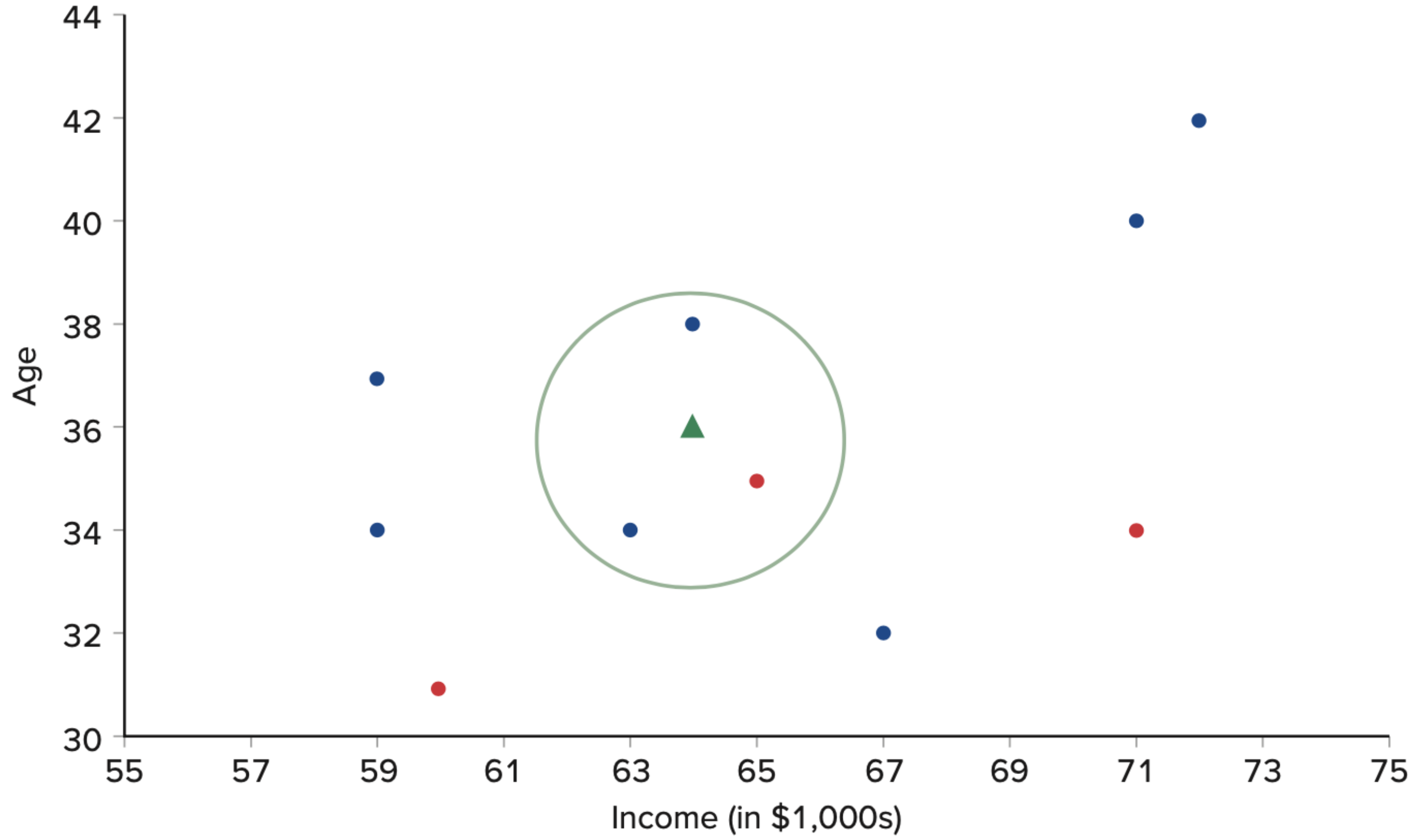
The *k*-Nearest Neighbors Method

- Example: Consider a mortgage company trying to determine whether or not a new applicant is likely to default on his or her home loan.
- The new applicant is 36 years old and has an annual income of $64,000.
- The mortgage company extracts information from 10 previous customers with similar income.

Customer ID	Income	Age	Default
001	65	35	Yes
002	59	34	No
003	71	40	No
004	67	32	No
005	60	31	Yes
006	59	37	No
007	72	42	No
008	63	34	No
009	64	38	No
010	71	34	Yes

The *k*-Nearest Neighbors Method

- A scatter plot of k-NN of k = 3, Age against Income for the 10 past customers
- The circle is for finding the prob. of default for a new customers with age 36, income 64,000



Performance Evaluation

- Performance measures for classification models can be computed from a **confusion matrix**

Actual Class	Predicted Class 1	Predicted Class 0
Class 1	No. of true positives (TP)	No. of false negatives (FN)
Class 0	No. of false positives (FP)	No. of true negatives (TN)

- Assume the success or target class is class 1 (e.g. detecting spam email)
- True positive: class 1 classified as class 1
  - Classify spam as spam and put into junk email folder
- True negative: class 0 classified as class 0
  - Classify good as good email and get it through.

$$\frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$

- False positive: class 0 incorrectly classified as class 1 [Type I error]
  - Classify good email as spam and put into junk folder.
- False negative: class 1 incorrectly classified as class 0 [Type II error]
  - Classify spam as good email

$$\frac{\text{False Negatives}}{\text{False Negatives} + \text{True Positives}}$$

Performance Evaluation

Actual Class	Predicted Class 1	Predicted Class 0
Class 1	No. of true positives (TP)	No. of false negatives (FN)
Class 0	No. of false positives (FP)	No. of true negatives (TN)

- Accuracy rate: proportion of observations that are classified correctly, $\frac{TP+TN}{TP+TN+FP+FN}$
- Misclassification rate: error rate, proportion of observations that are misclassified, $\frac{FP+FN}{TP+TN+FP+FN}$
- Sensitivity: (or Recall), the proportion of true positives among all actual positive cases, $\frac{TP}{TP+FN}$
- Precision: positive predicted value, proportion of true positives among all the predicted positive cases, $\frac{TP}{TP+FP}$
- Specificity: the proportion of true negatives among all actual negative cases, $\frac{TN}{TN+FP}$
- F1 score: Harmonic mean of Recall and Precision, which penalizes the lower score.

$$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Performance Evaluation

- Example: At FashionTech, Alissa has hired a social media marketing firm, MarketWiz, to develop predictive models that would help FashionTech acquire new customers as well as increase sales from existing customers.
- The confusion matrix where Class 1 (target) is the group of social media users who purchase at least one product after receiving a promotional message and Class 0 (nontarget) is the group of social media users who do not make a purchase after receiving a promotional message.

Actual Class	Predicted Class 1	Predicted Class 0
Class 1	29	19
Class 0	19	133

a. Calculate the accuracy rate
b. Calculate the sensitivity
c. Calculate the precision
d. Calculate the specificity

Performance Evaluation

Actual Class	Predicted Class 1	Predicted Class 0
Class 1	29	19
Class 0	19	133

a. Calculate and interpret the accuracy rate:

$$\frac{FP+FN}{TP+TN+FP+FN} = \frac{29+133}{29+133+19+19} = 0.81$$

b. Calculate the sensitivity: $\frac{TP}{TP+FN} = \frac{29}{29+19} = 0.604$

c. Calculate the precision: $\frac{TP}{TP+FP} = \frac{29}{29+19} = 0.604$

d. Calculate the specificity: $\frac{TN}{TN+FP} = \frac{133}{133+19} = 0.875$

Conclusion

- A concept of data mining and its siblings are introduced.
- Concept of Supervised and unsupervised Machine Learning (ML)
- kNN as an example of Supervised Learning is introduced.
  - A manual computation thru a sample case
- Performance evaluation for classification